MYP PERSONAL PROJECT REPORT

MAKING AN ONLINE ESL BOOK CLUB

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Objective A – Investigating

Ai) Goal

To design, code and publish an online book club that improves the English reading skills of ESL (English as a Second Language) students at St Andrew's Cathedral School.

I chose this goal as it coherently integrates my personal interests in reading and web design with my personal context as an ESL student. While drafting it, I utilised a goal development scaffold to ensure that it was *SMART* (see Appendix A).

The accessibility of websites, paired with the effectiveness of reading as a means of language acquisition, made them the natural means to achieve the target outcome of the project - the improvement of ESL students' English abilities. I decided that an online book club was the most logical fusing of these two concepts.

The challenge of this goal arose from my limited understanding of its content, as well as its interdisciplinarity. I had little knowledge of how reading could improve English ability, how comprehension could be taught, or how I could code and implement any of the features that I wanted to include in my website, which made the project quite intimidating. Furthermore, creating a product that was both functional and educational was challenging as concepts from each field had to be congruous with the other. The complex skills and high-level thinking required to achieve it meant that my goal was certainly challenging.

Ai) Global Context

Due to its high degree of engagement with the acquisition of language, my project is inextricably tied to the global context of Personal and Cultural Expression.

Language, in all of its forms, is the very means through which expression can take place. Without it, we would find it exceedingly difficult, if not impossible, to communicate. Take, for example, an exclusively Spanish-speaking person attempting to communicate with an exclusively English-speaking person via telephone. Without the aid of visual cues, such as body language, linguistic incongruities would make it immensely challenging for these people to meaningfully express information to one another. Language is essential to expression, and through my project, I aimed to help others express themselves by improving their linguistic capabilities.

Aii) Prior Learning – Web Design

Having a prior knowledge of web design was hugely beneficial for me as it was readily applicable throughout the project. Wireframing, a sketching technique for webpages that I learned in my Information and Software Technology course, was a skill that I applied to generate design ideas for each of my webpages (see Appendix B, Figures B2 & B3). This application of my prior learning streamlined my design process as it provided an effective

method of communicating ideas. Furthermore, its detailed representation of webpage layouts simplified the creation process by establishing clear, straightforward structures that allowed for a methodical approach to coding. My prior knowledge of web design facilitated the development of my product.

Aii) Prior Learning - Reading

As a former ESL student, I appreciated the frustrations that foreign language readers faced - for example, not understanding a word whose meaning was key to a sentence, losing track of a complicated plot, or being slow to read material that peers were able to breeze through. This experiential understanding of reading from the perspective of an ESL student meant that I could empathise with the target audience to make the product as relevant to them as possible. Part of my inspiration for a "vocabulary notebook" feature was my experience of not understanding new vocabulary - an experience shared by other ESL students, whose response to the feature was overwhelmingly positive. My empirical understanding of reading from the perspective of an ESL student gave me insight into the problems facing my target audience.

Aiii) Research Skills

During my research process, I sought a range of perspectives form multiple and varied sources, which was quite useful as it gave me a cornucopia of knowledge to draw from when developing the project. However, not all of the sources that I came across were wholly reliable or relevant to the project.

My research into book clubs, which aimed to identify the positive characteristics of book clubs that I could look to incorporate into my own project, yielded a prime example of source irrelevancy. I had found an article on book clubs by Esther Lombardi (2017), which only provided a cursory definition of the term. It was irrelevant to my research - I aimed to identify the value of book clubs, but as the article was purely informational, no value judgements could be made. Therefore, I couldn't find any positive aspects to include in my project. However, the source itself was reliable - Lombardi has reviewed and written literature for 20 years. Her experience makes her a reliable authority on book clubs, and books in general. From this research, I understood the fundamental definition of a book club.

On the contrary, some sources were highly relevant to my project. From my research into encouraging user interactions in an online educational space, which aimed to identify methods of improving user engagement with my project, I found a research paper by Banna et al (2015) that suggested improvements to online education forums based on data from an online course. The source's high degree of relevance made it very useful, as its information was directly applicable to my project. Furthermore, the source was quite reliable, as its authors' positions as professors at the University of Hawaii lends credibility to their conclusions. From this research, I identified strategies to promote user engagement with my project.

I utilised the OPVL source evaluation structure to thoroughly evaluate the relevance and reliability of the sources used during my research (see Appendix C).

Objective B - Planning

Bi) Success Criteria

I synthesised my research to establish a set of success criteria separated into relevant strands of factors affecting design. This provides an effective evaluation framework by clarifying the successful components of the project. These criteria can be found in Appendix D. These criteria were not changed throughout the project, as I was confident in the research that informed them, and believed that they were all achievable. I also wanted to preserve the criteria's authenticity - if a goal was not able to be achieved, I didn't modify it as it would interfere with the reliability of the evaluation.

Bii) Plan and Development

I developed detailed and logical short- and long-term plans with clear deadlines and records of change that describe the successful creation of my product, as well as a comprehensive account of my completion of this plan (see Appendix E, Tables E1 & E3).

Biii) Self-management Skills

Time and Task Management, and impact on Affective skills

To manage my project, I employed a range of time and task management strategies, utilising the self-assessment scaffold in Appendix G to assess my success.

When planning my research, I utilised the MISO brainstorming technique as a task management strategy (see Appendix E, Figure E1). An advantage of this technique is its encouragement of different source types, such as surveys or observations, which helped me to consider a greater variety of perspectives. Researching diverse perspectives improves information accuracy through the corroboration of different sources to eliminate bias.

However, not all of the research methods generated as a result of the MISO brainstorm were feasible. For example, one of my "Interview" research proposals was speaking to the parents of ESL students about their opinions of the program. However, I was unable to gain access to this target audience. I applied the affective skill of analysing and attributing causes for failure to identify a cause, and realised that I didn't have methods of contacting them - I didn't have their emails or phone numbers, nor any opportunities for face-to-face contact, as many of them were either overseas or didn't pick up their children from school. This made me realise the importance of accessibility when researching an audience.

Therefore, I also had to practice resilience by dealing with change, revising my interview questions for ESL students with in order to gain their perspectives on the ESL program

instead. This meant that I was still able to conduct valuable primary research about the topics relevant to my project.

I am a Practitioner in self-management - I can readily demonstrate methods of managing my time and tasks, but these methods are not always successful in keeping me on track.

Reflection skills: Keep a journal to record reflections
I was a strong journal writer at the start of the project due to my familiarity with its process. I have written reflection logs in my Information and Software Technology and Design courses, and have independently developed a structure for writing them. As I can independently write reflective journals using my own scaffold, I am an Expert in keeping a journal to record reflections.

Organisation skills: Use appropriate strategies for organising complex information
As part of my research process, I summarised information from a number of complex research papers. I extensively utilised the Cornell note taking method to organise the information in these papers, which allowed me to identify areas of comprehension and confusion, and helped me to succinctly collate information (see Appendix H). My ability to independently organise complex information using an appropriate strategy means that I am an Expert in this skill.

Objective C - Taking Action

Ci) Goal Achievement

After 35 hours of work, I published ReadingAid.site. To evaluate its success, I referred to my goal:

To design, code and publish an online book club that improves the English reading skills of ESL students at St Andrew's Cathedral School.

The first component of the goal was "to design, code and publish an online book club." During development, I generated a webmap showing the structure of the website as well as a wireframe sketch for each page (see Appendix B, Figures B1, B2 & B3). From these designs, I coded and published the book club, which gained positive feedback from 88% of ESL student survey respondents from the school. The practicability of the designs, as well as their successful implementations, meant that I achieved the first component of the goal.

The second part of the goal, however, was not achieved through the project. The online book club didn't improve the English reading skills of the school's ESL students - after my presentation, most students did not utilise the site again, and from observation of their reading habits, they did not implement the reading strategies from the website (such as using a vocabulary notebook or "previewing" the text). The English reading skills of students at the school did not improve after using the online book club.

The goal was partially achieved. While I was able to design, code and publish an online book club, the product did not achieve its outcome thoroughly enough to wholly satisfy the goal.

Global Context
My project was inextricably linked to the Global Context of Personal and Cultural Expression. Through its focus on reading, the project was inherently related to the discovery of new ideas and cultures. In my research into effective methods of language acquisition through reading (see Appendix H), I investigated tools with which this discovery process could be facilitated, and by surveying ESL teacher' opinions of reading (see Appendix I, Figure I3), I explored the perspectives of those involved with it. The project's focus on reading helped ESL students to discover new ideas and cultures, while allowing me to explore theirs. My project was connected to the global context of Personal and Cultural Expression through its exploration of the ways in which new ideas and cultures are discovered.

Cii) Thinking skills

I applied a variety of creative and critical thinking skills relevant to the project, as well as transferred learning from my research, in order to achieve my goal.

Creative thinking: Use brainstorming and visual diagrams to generate new ideas or enquiries
I used brainstorming as a method of idea generation for my project - I identified my areas of interest, and came up with ideas that could incorporate them. Brainstorming allowed for the rapid generation of new enquiries, as it encouraged quantity over quality of ideas. Furthermore, its fluid representation of ideas meant I could think flexibly, connecting them in a variety of ways to generate new ones. However, the limited feasibility of the ideas generated from brainstorming restricted its effectiveness. Nonetheless, it was still powerful method of prompting creative thought.

Critical thinking: Apply skills and knowledge in unfamiliar situations
While coding the vocabulary notebook, I applied my knowledge of one-to-many data models to implement its functions. Even though I had no experience writing code for an online "notebook," the application of my knowledge meant that I created a solution that functioned as intended. This process was quite tedious, however, as my unfamiliarity with the situation meant that creating the solution involved a lot of trial and error.

Transfer: Combine knowledge, understanding and skills to create products or solutions
In the "individual title" page, my knowledge of effective language acquisition methods (vocabulary notebook), strategies to engage users in an online educational system (discussion), strategies to encourage engagement with text (preview and notes) and my skills in using MySQL, PHP, JavaScript, HTML and CSS were ultimately combined to create a novel, multi-faceted webpage that engaged the user in reading - 100% of survey

respondents stated that it was "helpful" in engaging them with the book (see Appendix I, Figure I1).

I am a Practitioner in applying thinking skills - I am familiar with, and can comfortably utilise, a range of skills and strategies to manage my thinking, however, these skills are not always effective in achieving their outcome.

Ciii) Communication and social skills

Communication skill: Use a variety of media to communicate with a range of audiences
Communicating with my audience was central to my project - my ongoing interaction with the audience informed the creation of the product, improving its relevance to them. After verbal communication with the school's ESL coordinator, justifying how my product could aid ESL students' reading skills, I gave a multimedia presentation to them in which I introduced them to the site and asked them to complete a survey about it. I also designed a survey specifically for the ESL coordinator to gain a teacher's perspective of the site. Communication - whether verbal, through surveys or multimedia presentations - helped me to develop a deeper understanding of my audience.

Social skill: Listen actively to other perspectives and ideas
I received a particularly interesting survey response suggesting a Chinese book to add to the site. While I was initially dismissive of this idea, I considered the respondent's perspective more actively, and realised that they meant an English translation of the book. Featuring books with content and contexts familiar to the reader would be helpful in engaging them with the text, an idea that I had not previously considered. By listening actively to other perspectives and ideas, I could make the product more suitable for the audience.

I am a Practitioner in applying communication and social skills. I am able to communicate with a variety of audiences in a variety of methods, however, I can be quick to dismiss the ideas of others without careful consideration.

Objective D – Reflecting

To evaluate the success of my product, I matched it against the set of success criteria that I established. This can be seen in Appendix D.

Di) Evaluation

By dividing the success criteria into specific strands, I could identify the aspects in which the product was successful. Moreover, by establishing a rigid set of criteria and measuring quantitatively, I could gain an accurate and insightful understanding of the its success.

The website met most Aesthetics and Appropriateness criteria, and around half of the Function and Accessibility criteria. The high Appropriateness score suggests that the

solution thoroughly considered the target audience in order to be suitable for them. The high Aesthetics score shows that I was able to successfully execute the aesthetic specifications outlined in the criteria. However, negative feedback about whitespace on the website suggests that users would prefer for content to be more spread out. Furthermore, these specifications were not too technically demanding - nonetheless, I believe that the product was of an acceptable aesthetic quality.

The lower Function score can be attributed to limited technical proficiency. Although I had learned plenty of web design content, I was still not able to successfully code certain features stipulated in the criteria. In terms of Accessibility, the project was able to satisfy most of the audience-determined criteria - however some others, such as the website being among the first results of a Google search, were unachievable due to a lack of thorough research and monetary investment.

Di) Possible Improvements

As the final product did not completely satisfy the success criterion, improvements could be made to enhance its quality:

Function

Success Criterion Cii) Click and Clunk - The website could implement a "click and clunk" feature to aid users with comprehension (Klingner & Vaughn, 1998). This could be achieved by allowing users to highlight sections of text that "click," or make sense, as well as "clunks" (sections that don't). This would allow for independent user identification and resolution of areas of weakness, thereby improving comprehension skills.

Accessibility

Success Criterion Ai) Search Engine Optimisation - The website was not one of the top results when the term "online ESL book club" were searched by Google - in fact, it was not among the first 176 results listed. This limits its ability to attract new users, as websites are mostly accessed through search engines, and potential users will not come across the website from a search. In order to improve this, a ".com" domain could be purchased to increase the likelihood of the page being featured as a search result. Keyword optimisation software, such as WordStream, could also be used to identify keywords that best promote the website to search engines (Hardwick, 2018).

Dii) Growth in Knowledge and Understanding

The global context that I grounded my project in was that of Personal and Cultural Expression. Throughout the course of the project, my understanding of the significance of this global context has certainly deepened. Initially, I justified the its relevance to the project by linking them through language - I stated that, as self-expression was impossible without language, helping others to acquire a new language therefore helped them to express themselves. As I learned new programming languages as part of my skill acquisition process, I realised that language not only helped people to express themselves,

but that it also changed how people express themselves. By learning new programming languages, I discovered how to communicate information to my computer so that it would in turn divulge knowledge that helps others to develop a skill. Through code, I could not only communicate ideas: I could also communicate a sense of community, comprehension skills, and a passion for reading, all of which would be difficult to express through plain verbal or textual communication.

Furthermore, I found that languages could also be used in conjunction to enhance meaning, in that body language can be used in conjunction with a spoken language to add depth to expression. This insight was inspired by the role that each programming language played in the creation of my product - HTML & CSS, for example, were responsible for styling webpages, while PHP was responsible for retrieving and displaying information from the database. Personal expression is deepened through the use of multiple languages.

Through the completion of my project, I have been able to gain a broader knowledge of Personal and Cultural Expression.

Diii) Development as a Learner

Completing the Personal Project has certainly seen me develop as a learner - because it was so challenging for me, I had to rely on my strengths to identify and improve upon my weaknesses. For me, the areas that saw the most growth were my inquiry skills and reflectiveness.

Being a passionate Inquirer was vital during my research process. Due to the project's ambitious goal, I had to conduct a large volume of interdisciplinary research just to be sufficiently informed to create the product. During my research into reading strategies and online user engagement, the complexity of my sources, as well as the tedious summary process, meant that I was not initially motivated to work. However, by breaking research pieces into smaller sections and making an effort to engage with the content, my research soon became manageable, and even interesting. This inquiring mindset carried over to my studies, and helped me to develop a genuine passion for learning. Through developing my research skills, I was able to become a better Inquirer.

At the start of the project, reflectiveness was not one of my strengths. I usually considered my creative process to be "complete" with the creation of a product, and would rarely reflect on work independently. This meant that I had limited understanding of my skills as a learner, making learning more difficult. During this project, I reflected as much as possible, writing a log at the end of each session of work. Practicing reflectiveness meant that I could keep track of completed work and identify areas for improvement, which ultimately made it more effective. It is a skill I can apply outside of the project in order to further my growth as a learner.

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Appendix A: Goal Development with SMART scaffold

GOAL: To design, code and publish an online book club that improves the English reading skills of ESL students at St Andrew's Cathedral School. This goal is SMART: Specific, Meaningful, Attainable, Realistic, and Time-bound.

Specific: The goal specifies the target audience of the project, the impact that the project should have on the target audience, the medium through which the outcome of the project will be achieved, and ties in with the global context of Fairness and Development through its focus on giving fair opportunities to immigrants to English-speaking countries by improving their ability to communicate.

Meaningful: The goal is personally relevant to me as I am a first-generation immigrant to Australia who has struggled to learn English, and who has also been a part of an ESL program. The accomplishment of my goal means that the immigration experience for people in circumstances like mine would be made easier.

Attainable: The goal specifies that I must "design, code and publish an online book club." This is something that I feel will be attainable, given my experience with designing, coding and publishing webpages. However, the "book club" aspect of the goal could be quite challenging for me, as I have no experience in creating one as an online medium, but with the help of online resources, I should be able to accomplish the goal.

Realistic: Given the resources currently available to me, I should be able to complete the project - I have access to a laptop capable of designing, coding and publishing a website, the educational resources that I will need to learn those skills, and the research skills necessary to investigate relevant topics to the project. I am confident in my ability to accomplish the goal.

Time-bound: The project should not take longer to complete than the due date – according to Eric Baum of marketing firm Bluleadz (2015), the average website takes around 30 days complete from start to finish, and I have set aside about 60 days to complete this project.

Appendix B: Website Design Ideas

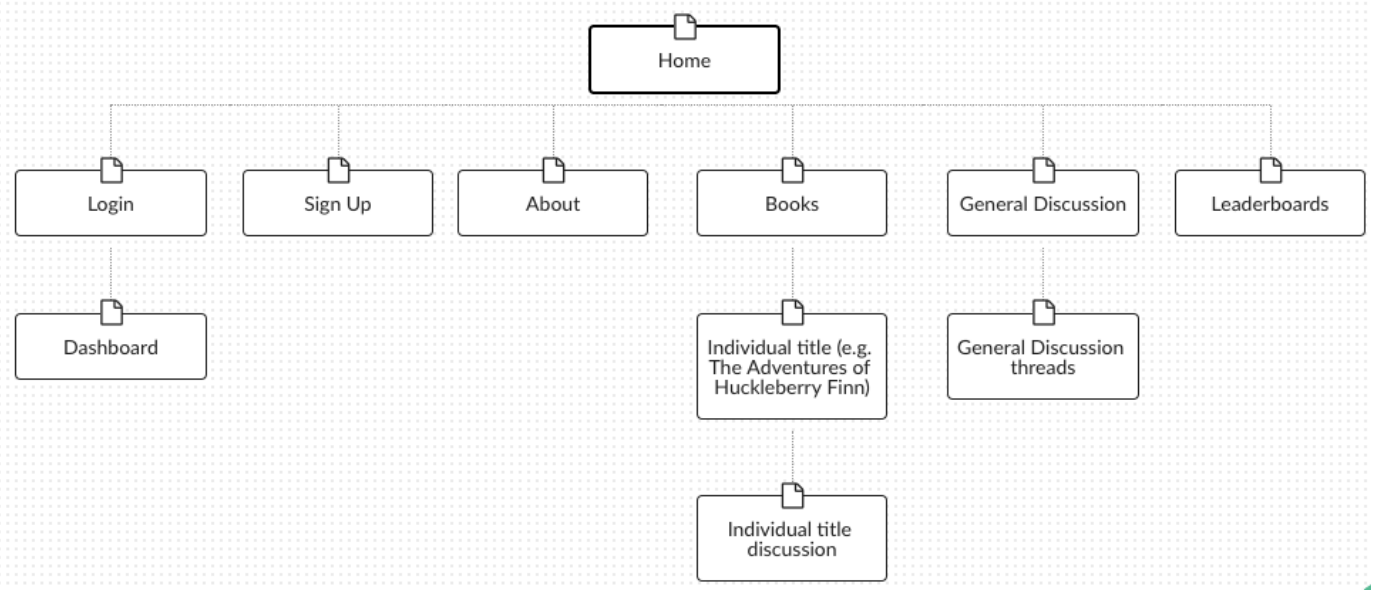


Figure B1. Webmap structure

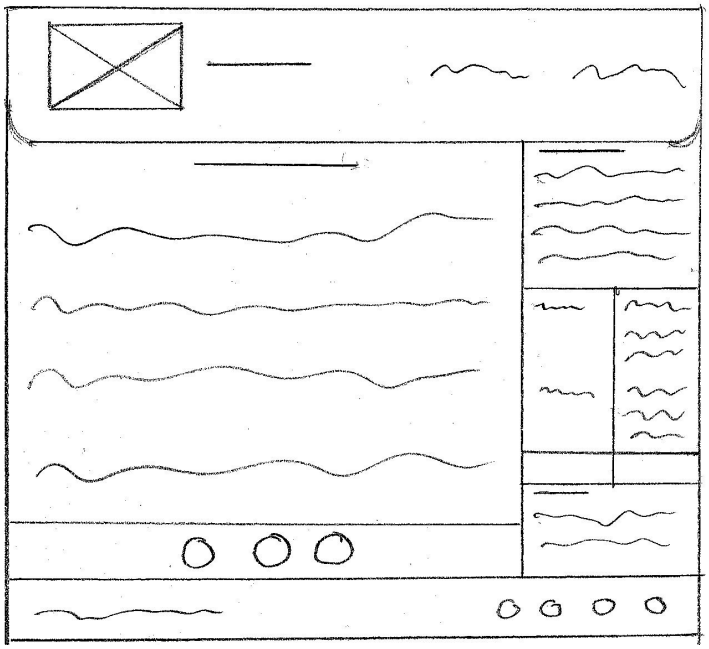


Figure B2. Wireframe sketch for Home Page

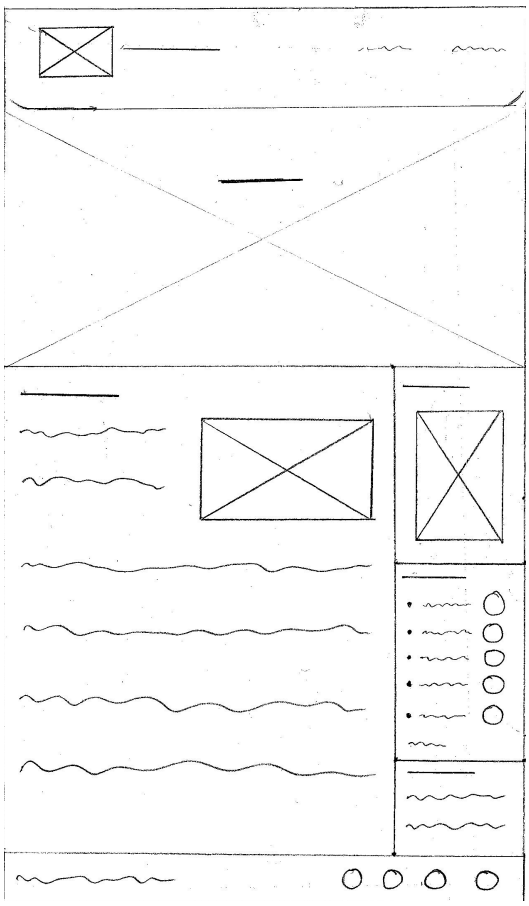


Figure B3. Wireframe Sketch for Individual Title Page

Appendix C: OPVL Source Analysis

Source	Origin	Purpose	Values	Limitations
The Benefits of Extensive Reading (ER), by Richard R. Day	Richard R. Day was the creator of this secondary source, which was published by Oxford University Press. Day is a professor of ESL at the University of Hawaii, which lends credibility and reliability to this source.	Day most likely created this paper to encourage ESL educators to promote and enforce an extensive reading program in their own ESL courses.	The source provides valuable information on the positives of regular reading, and is also relevant to this piece of research as it addresses and evaluates the effectiveness of reading a means of language acquisition.	It is limited by its lack of detail in terms of how to encourage ESL students to read widely, merely citing their benefits.
Giving Readers a Voice: Book Discussion Groups by Anna Healy	This secondary source was created and published by librarian Anna Healy in 2002. As a librarian, Healy has credibility, as she has successfully started and managed a book club of her own. This makes the source more reliable.	Healy's intention in creating this piece of work is to inform the reader of how to run a book club. The author's purpose in creating this source closely matches the purpose of the research, making it highly relevant.	In this piece, Healy's support of book clubs is made clear through the fact that she lists their advantages in order to promote them.	Healy does not address the disadvantages or limitations of book clubs, possibly due to her bias in their favour.
Interaction matters: Strategies to promote engaged learning in an online introductory nutrition course by Banna et al	The secondary source was created by Jinan Banna, Meng-Fen Grace Lin and their associate professors at the University of Hawaii. It was published by the MERLOT Journal of Online Learning and Teaching in 2015.	The document exists primarily to suggest improvements that can be made in the online education forum using evidence gathered from a study done with an introductory nutrition course. It is highly relevant to this piece of research as its purpose is very similar.	The source is reliable, as Banna and Lin's positions as professors at the University of Hawaii means that they have quite a lot of authority when it comes to this field due to their qualifications. Furthermore, they collected primary evidence to back up their claims.	A limitation of this document is that its findings are taken from one set of data, meaning that they can be inconsistent with data found from other sources, reducing the validity of its conclusions.

Appendix D: Evaluated Success Criteria

Strand	Success Criteria (Relevant Research)
Aesthetics: **3/4**	• Ai) The website must use blue as its primary colour (Existing products analysis) - Criterion met: The majority of the website's pages use blue in several places throughout their designs. • Aii) The website must utilise a colour palette generated by the Adobe Kuler tool (Grok Learning, n.d.) - Criterion met: The colour theme used for the site was created by Kuler. • Bi) At least 70% of users must be satisfied with the amount of whitespace used on the website (Grok Learning, n.d.) – Criterion not met: According to a survey of ESL students, 56% of users were satisfied with the amount of white space on the website, with the other 44% reporting that they were somewhat satisfied (See Appendix I, Figure I2. • Bii) Paragraph fonts must be sans-serif (Grok Learning, n.d.) - Criterion met: The website uses sans-serif fonts almost exclusively.
Function: **4/7**	• Ai) Within the initial week of being introduced to the website, at least 70% of users must read a book in any format 3 or more times (Day, n.d.) – Criterion not met: according to Google Analytics, only 10% of users accessed the site to read a book. •Bi) The product must include a public "discussion" section, where users can communicate textually to each other, that features both academic and non-academic discussion threads (Banna et al, 2015) - Criterion met: There is a "General Discussion" page on the website. • Bii) The product must allow for users to post anonymously on all of its discussion threads (Banna et al, 2015)- Criterion met: Users can choose to post anonymously on any discussion thread by selecting the "I'd rather post as a guest" option on their comment. • Ci) The product must include a "preview" feature for all of its featured books that instructs users on how to preview the text (skim the text to make a prediction) (Klingner & Vaughn, 1998) - Criterion met: There is a short preview feature for featured books, which includes a synopsis and a statement instructing them to skim the text and make a prediction about it. • Cii) The product must include a "click and clunk" feature for all of its featured books that instructs users on how to use the "click & clunk method" when reading the text (Klingner & Vaughn, 1998) - Criterion not met: The "clink and cluck" feature was unable to be included on the site as it required a higher level of technical proficiency to execute than I had at my disposal. • Ciii) The product must include a "summary" feature for all of its featured books that instructs readers on how to summarise the text (Klingner & Vaughn, 1998) - Criterion not met: Users can only take notes about the text, and are given no scaffolding to summarise it. • Civ)The product must include a "vocabulary notebook" feature that allows users to record new vocabulary and their definitions (Oxford Online English, 2017) - Criterion met: Users can create a list of words and their definitions, and keep track of those words as they read.

Accessibil-ity: 2/4	• Ai) The website must be one of the top 20 search results returned when the keywords "online book club" or "online ESL book club" are searched on Google (Hardwick, 2018) - Criterion not met: The webpage cannot be found within the first 18 pages of results (roughly 176 results). • Bi) At least 70% of users must be satisfied with how clearly the discussion section's threads are displayed (Banna et al, 2015) - Criterion met: 80% of survey respondents could access the General Discussion threads, with some describing it as "formal" (See Appendix I, Figure I1) • Bii) 100% of users must be able to access and utilise all the tools and functions of the website without guidance (Keep et al, 2016) – Criterion not met: Only 70% of survey respondents were able to navigate the site entirely, and utilise all of its tools and functions. • Ci) The page must be able to be translated into the 20 most common languages (Existing Products analysis) - Criterion met: The web page can be translated into more than 100 different languages.
Appropriate-ness: 6/7	• Ai) A "book poll" where users vote for featured titles must be included (Jacobsohn, 1998) - Criterion met: On the home page, there is a poll in which users can vote for a title to be added in the following month. • Aii) Each user of the site must be able to sign up for and access an individual profile (Banna et al, 2015) - Criterion met: Users can use the Log In and Sign Up pages to create and view individual profiles. • Aiii) The moderator(s) of the website must log onto the website daily, and post on its forums at least three times a week (Banna et al, 2015) - Criterion not met: the website was monitored on a daily basis, and comments were not regularly made by its moderators. • Bi) Any individual activity on the site must be able to be completed by the user in less than 30 minutes (Banna et al, 2015) - Criterion met: The site's preview feature, the only reading-specific activity available on the site, is completed within 2-3 minutes. • Bii) Information about a book must be made available to the user before they read it (Texas Education Agency, n.d.) - Criterion met: On the Book List Page, a short description of each available title, as well as its author and date of publication, is shown to the user. • Ci) The website must implement strategies such as copyright and trademark to protect its intellectual property (Australian Copyright Council, 2017) - Criterion met: The footer of the site displays its year of copyright, and acknowledges that any external intellectual property displayed on the site belongs to its respective owners. • Cii) Books that are featured or published on the site must (Australian Copyright Council, 2017) - Criterion met: All of the available titles on the site are public domain books whose copyrights have expired.

Appendix E: Planning Methods

Action	Estimated length & deadline (actual date)	Notes
1. Identifying and acquiring the skills and knowledge necessary to the creation of the product	Start: 31/12/18 (31/12/18) Finish: 19/1/19 (9/2/19)	Revise content such as HTML, CSS, JavaScript, PHP & mySQL.
2. Establishing the hierarchy of the website and designing the layout and appearance of each individual site	Start: 20/1/19 (10/2/19) Finish: 26/1/19 (23/2/19)	Draw designs for the sites using wireframe sketches – research to be done into this!!
3. Coding the website – front-end (HTML & CSS), middleware (JavaScript) and back-end (PHP & mySQL)	Start: 27/1/19 (24/2/19) Finish: 9/2/19 (28/4/19)	Deadlines probably subject to change, depending on scheduling of assessments during the term
4. Publishing the product	Start: 10/2/19 (28/4/19) Finish: 11/2/19 (28/4/19)	Use reliable web hosting services such as BlueHost or GoDaddy – do research into prospective web hosting sites
5. Testing the effectiveness of the product in improving the English reading skills of ESL students	Start: 12/2/19 (29/4/19) Finish: 19/2/19 (7/5/19)	Make contact with ESL department at SACS and ask students to use the website for a one-week period

Table E1. Long-Term Plan

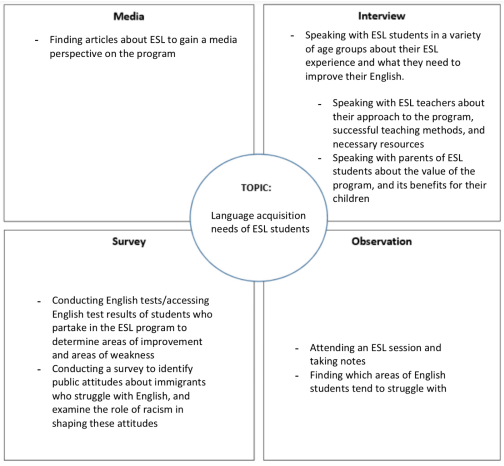


Figure E1. MISO Research Plan

Procedural Steps - Plan	Resources Needed	Deadline	Changes Made
1. Revise prior knowledge of HTML & CSS by completing the Grok Learning "Introduction to HTML & CSS" course, as well as reading through and completing exercises in the book "Head First HTML & CSS"	Computer, access to internet, Grok Learning account, copy of "Head First HTML & CSS" (obtainable online via the O'Reilly Media database)	5/1/19	
2. Learn JavaScript by reading through and completing exercises in the book "Head First JavaScript Programming"	Computer, copy of "Head First JavaScript Programming" (obtainable online via the O'Reilly Media database)	12/1/19 Met on 13/1/19	I took longer than planned with the Head First JavaScript book, as I just did not have sufficient time to fully understand the contents of the book. I then readjusted my deadlines to account for that extra day.
3. Learn PHP & mySQL by reading through and completing the exercises in "Head First PHP & mySQL," and completing the Introduction to Databases (SQL) course on Grok Learning	Computer, access to internet, Grok Learning account, copy of "Head First PHP & mySQL" (obtainable online via the O'Reilly Media database)	19/1/19 Met on 9/2/19	The PHP book was especially difficult for me, as it covered content that I was not familiar with. This meant that I hadn't finished the book when school started, which further slowed my progress. I then adjusted my deadlines accordingly.
4. Establish and create a visual representation of the website hierarchy (or "sitemap") using the sitemap tool Writemaps.	Computer, access to internet, Writemaps trial account, pencil, paper	22/1/19 Met on 20/2/19	
5. Design the layout and appearance of each individual site on the website with wireframe sketches	Pencil, paper, ruler, eraser	26/1/19 Met on 23/2/19	
6. Code the home and about pages for the website in Brackets, using the design created in step 5 as a reference. Use Adobe Kuler to generate a colour theme for the website.* *step added 24/2/19	Computer, access to internet, Brackets software (coding interface), copies of "Head First PHP & mySQL," "Head First HTML & CSS," and "Head First JavaScript Programming"(obtainable online via the O'Reilly Media database)	2/2/19 Met on 17/4/19	This step took me a long time to complete as its original deadline conflicted with the start of my school's assessment period, when I had a lot of other work to do. To cope with this, I delayed the deadlines once more.
7. Code the login and sign up pages for the website in Brackets, using the design created in step 5 as a reference.* *step added 24/2/19	Computer, access to internet, Brackets software (coding interface), copies of "Head First PHP & mySQL," "Head First HTML & CSS," and "Head First JavaScript Programming"(obtainable online via the O'Reilly Media database)	2/2/19 Met on 20/4/19	
8. Code the books and leaderboards pages for the website in Brackets, using the design created in step 5 as a reference.* *step added 24/2/19	Computer, access to internet, Brackets software (coding interface), copies of "Head First PHP & mySQL," "Head First HTML & CSS," and "Head First JavaScript Programming"(obtainable online via the O'Reilly Media database)	2/2/19 Met on 22/4/19	
9. Code the general discussion, general discussion threads and individual title threads for the website in Brackets, using the design created in step 5 as a reference.* *step added 24/2/19	Computer, access to internet, Brackets software (coding interface), copies of "Head First PHP & mySQL," "Head First HTML & CSS," and "Head First JavaScript Programming"(obtainable online via the O'Reilly Media database)	9/2/19 Met on 24/4/19	I merged the general discussion and its threads together from two separate pages, as separating the features would significantly complicate the coding process.
10. Code the dashboard and individual title pages for the website in Brackets, using the design created in step 5 as a reference.* *step added 24/2/19	Computer, access to internet, Brackets software (coding interface), copies of "Head First PHP & mySQL," "Head First HTML & CSS," and "Head First JavaScript Programming"(obtainable online via the O'Reilly Media database)	9/2/19 Met on 28/4/19	
11. Publish the website to the internet through a website hosting service	Computer, access to internet, website domain, website hosting service	9/2/19 Met on 28/4/19	Not exactly a change, but I used the iPage website hosting service to publish my product.
12. Contact ESL teachers at SACS to discuss the possibility of introducing students to the product	Computer, access to internet, email account	11/2/19 Met on 3/5/19	
13. Attend an ESL session, and introduce students to the product. Give them access to the site, and conduct a survey on its function, aesthetics, appropriateness and accessibility (as per its success criteria). Track their use of the product over the period of one week with website tracking software e.g. Google Analytics	Website, Google Analytics registration, survey	19/2/19 Met on 7/5/19	

Table E2. Short-Term Plan

Appendix F: Development Logs

Objective 1: Learning HTML & CSS (31/12/18 - 5/1/19)
In this part of the creation process, I sought to polish my skills in HTML & CSS, the simplest web programming languages, and the ones that I was most familiar with. I did this by first completing an "Introduction to HTML/CSS" course on Grok Learning in order to refresh my knowledge of the languages. Next, I found the book "Head First HTML & CSS" on the Safari Books Online website, and read through it, completing its exercises and mini-projects. I chose the Head First books to learn from as I had read Head First Java, and enjoyed the ways in which the books taught programming. This book covered the "flow" of a webpage, the elements that could be implemented as part of it, and the range of webpage design styles. After a week of work on the book (split into roughly 3-hour sessions), I had completed it by the deadline.

Objective 3: Learning PHP & MySQL (14/1/19 - 9/2/19)
To complete this objective, I read through and completed exercises from "Head First PHP & MySQL." This was by far the most difficult book for me to complete, as I had no experience with these languages, or with the databases that they relied upon to work. Furthermore, while I was halfway through the book, I had to go back to school, which took up a lot of time and slowed down my ability to make progress with it. The book covered topics such as inserting and retrieving database records, browser cookies, and PHP expressions. I then completed the "Introduction to Databases (SQL)" course on Grok Learning quite easily, as it assessed content that I already had from the book.

Objective 7: Login and Sign Up Pages (18/4/19 - 20/4/19)
The log in and sign up pages of the website were created to allow users to track their individual progress and development. By following the wireframe sketch and using the navigation bar template from the home page, I was able to style the page quite easily using CSS. I used a "Jello" layout for this site (main content is kept centred with a fixed width) as I wanted to have centred Login and Sign Up panels. Code-wise, these pages mostly involved inserting and retrieving information from a database, something I could do using PHP.

Objective 12: Contact ESL teachers (29/4/19 - 3/5/19)
To achieve this objective, I went to my school's ESL coordinator, Mrs Burke, in order to discuss the possibility of me presenting the website to a group of ESL students, and getting feedback from them in the form of a survey. My face-to-face negotiations were successful, and I was able to secure a presentation opportunity at the start of an after-school ESL homework club. I then prepared a survey for the students to complete.

Objective 13: Present website and collect feedback (4/5/19 - 7/5/19)
I presented my website to a group of 10 ESL students at my school who were in years 10-12, as well as the ESL Coordinator, at the start of an after-school ESL homework club. I talked to them about my project and the goal that I hoped to achieve with my website, and spoke about my own experiences as an ESL student and the way that they shaped my decision to do this project. I then showed them some of the site's features, and allowed them to explore the site on their own devices as they completed my survey.

Appendix G: Self-Evaluation Rubric

Self Assessment of Learning Skills

Novice ***Watch***	**Learner** ***Copy***	**Practitioner** ***Do***	**Expert** ***Teach***
Can watch others performing tasks and using the skill High levels of scaffolding from teacher needed	Can copy others performance of the skill Medium level of scaffolding needed	Can demonstrate the skill on demand Minimal teacher scaffolding required	Can teach others the skill No teacher scaffolding required

Appendix H: Research Summary - Improving English Comprehension Skills

The list below details some of the methods that can be used to improve the effectiveness of reading to develop English comprehension skills:

1. Read regularly: Like any skill, reading can only be improved through a process of repetition and practice. According to a paper written by Richard R. Day PhD., professor of ESL at the University of Hawaii, the benefits of extensive reading (or "free reading") are numerous - studies done on extensive reading in ESL and EFL (English as a Foreign Language) groups show that, across all ages and around the world, it leads to increases in vocabulary; reading, writing, and speaking proficiency; reading motivation and attitude; and reading rate (a key effector of reading fluency) (Day, n.d.).

2. Use a vocabulary notebook: One of the major benefits of reading is that it can be a very effective way to expand the reader's vocabulary. In order to maximise this benefit, it is recommended that the reader keeps a "vocabulary notebook"- a record of new vocabulary (Oxford Online English, 2017). This can help readers move new vocabulary from their short term to their long term through spaced repetition (Oxford Online English, 2017). These notebooks can help readers to rapidly expand their vocabulary, which is not only beneficial for reading English, but also for writing, listening and speaking it (Oxford Online English, 2017).

3. Pick the right book: An integral part of the development of the reader's skill is challenge. In order to develop their reading ability, Geikhman (n.d.) states that the reader should strive to read books that are just beyond their "comfort zone." Additionally, the reader should attempt to read books in genres that they enjoy, as this will provide motivation to persevere with the text (Geikhman, n.d.).

Appendix I: Survey Results

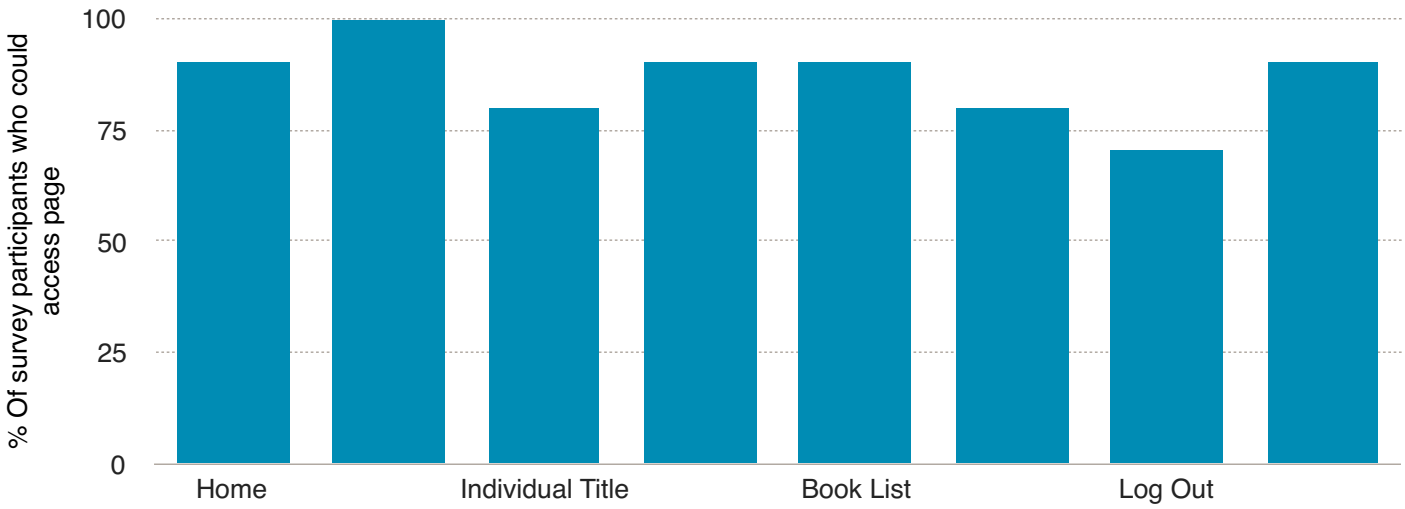


Figure I1. Accessibility Survey Results

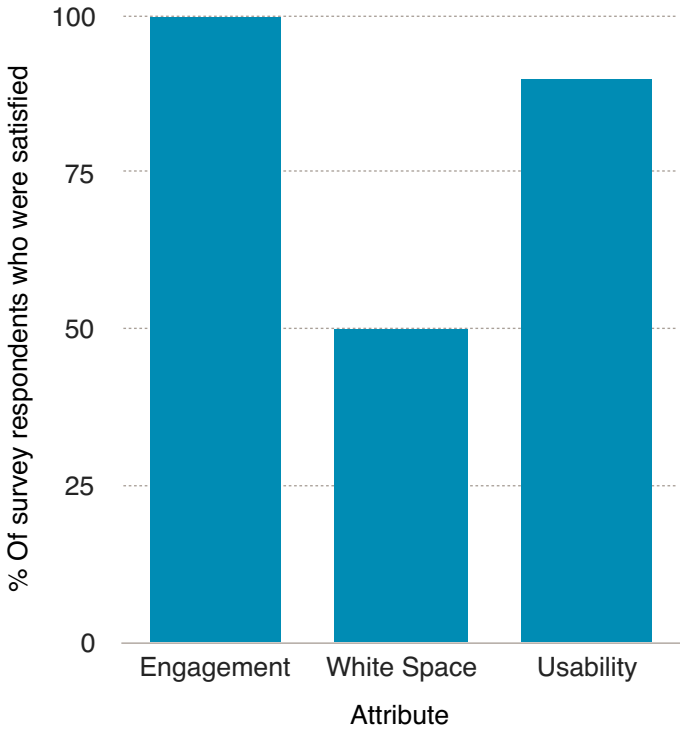
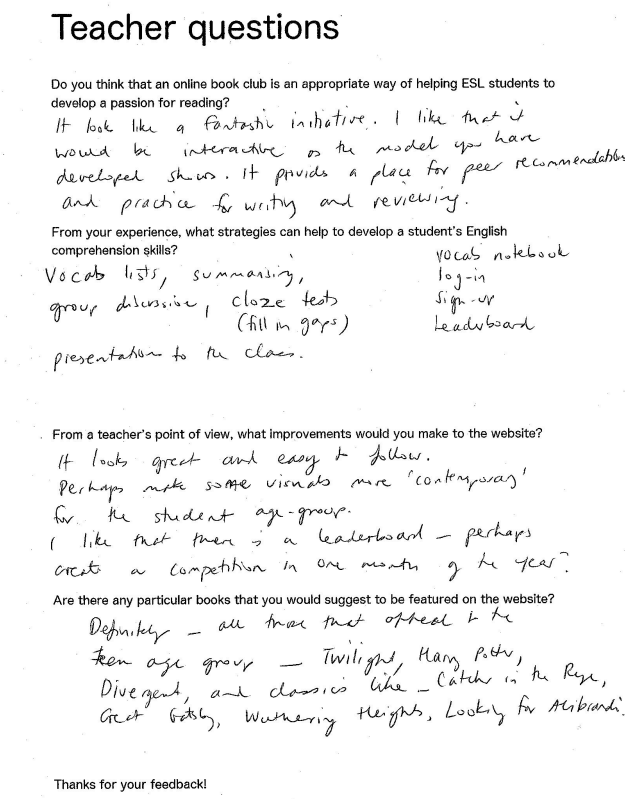


Figure I2. Website Satisfaction Survey Results



Teacher questions

Do you think that an online book club is an appropriate way of helping ESL students to develop a passion for reading?

It look like a fantastic initiative. I like that it would be interactive as the model you have developed shows. It provids a place for peer recommendations and practice for writing and reviewing.

From your experience, what strategies can help to develop a student's English comprehension skills?

Vocab lists, summarising, group discussion, cloze tests (fill in gaps) presentation to the class.

vocab notebook
log-in
sign-up
Leadrboard

From a teacher's point of view, what improvements would you make to the website?

It looks great and easy to follow. Perhaps make some visuals more 'contemporary' for the student age-group. I like that there is a leaderboard – perhaps create a competition in one months of the year?

Are there any particular books that you would suggest to be featured on the website?

Definitely – all those that appeal to the teen age group – Twilight, Harry Potter, Divergent, and classics like – Catcher in the Rye, Great Gatsby, Wuthering Heights, Looking for Alibrandi.

Thanks for your feedback!

Figure I3. Teacher Survey Results